

E-COMMERCE DATA ANALYSIS

Revenue, Profitability & Customer Behavior Report

End-to-End Analysis using Python, MySQL, and Power BI

Prepared for: E-Commerce Business Stakeholders

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1. Executive Summary

This report presents an end-to-end data analysis project built around an e-commerce transactional dataset spanning customer profiles, product catalog information, transaction records, browsing sessions, and product reviews. The objective of the project was to understand what drives revenue, why growth has been inconsistent across months and categories, how order cancellations affect profitability, and which products, brands, and customer segments matter most to the business.

The analysis was carried out in three connected stages. First, the raw datasets (customers, products, transactions, sessions, and reviews) were cleaned, merged, and enriched using Python (pandas, NumPy, Matplotlib, and Seaborn) inside a Jupyter Notebook. Second, the cleaned dataset was loaded into a MySQL database (E_COMMERCE_BEHAVIOR) and queried with structured SQL to answer fourteen specific business questions covering revenue, customer behavior, product performance, and discounting patterns. Third, the transaction-level data was modeled in Power BI to build an interactive dashboard that tracks total revenue, profit, order volume, quantity sold, and revenue growth, sliceable by category, year, and order status.

Across all three stages, a consistent picture emerges: Electronics is the single largest revenue-generating category (roughly a quarter to just over a quarter of total revenue depending on the view), a small number of brands and products account for a disproportionate share of sales, discounting has a non-linear and sometimes counter-productive relationship with revenue, and cancelled orders represent a material and avoidable drag on profitability. Revenue is also highly seasonal, peaking toward the fourth quarter of the year and dipping in February. These findings, together with detailed recommendations, are presented in full in the sections that follow.

2. Business Problem

The e-commerce company at the center of this analysis has been experiencing inconsistent revenue growth across months and product categories. Some periods show healthy growth while others show outright negative growth, making forecasting and inventory planning difficult. In addition, a meaningful share of orders are being cancelled after they are placed, which erodes profitability even when top-line revenue and order counts look healthy. Orders and quantity sold also do not always move in the same direction as profit, which suggests that volume alone is not a reliable indicator of business health.

The specific pain points that motivated this project were:

- Revenue fluctuates significantly from month to month, with no clear, communicated explanation for the swings.
- Certain months show negative growth compared to the prior period, creating uncertainty for planning and budgeting.
- The Electronics category contributes the highest share of revenue, but it is not clear whether other categories are being under-leveraged.
- Cancelled orders are reducing realized profit, even though they still consume operational resources (processing, packaging, logistics).
- Orders and quantity figures do not always align with profit, indicating that revenue and profit need to be tracked as separate, complementary metrics.

3. Business Objectives

Based on the business problem above, the project was scoped around five concrete objectives:

- Improve the consistency of revenue growth by identifying the drivers behind high- and low-performing periods.
- Identify which months and periods over- and under-perform, to support more accurate forecasting and staffing/inventory decisions.
- Optimize product category performance by understanding which categories, products, and brands generate the most revenue and demand.
- Reduce order cancellations by understanding where and why they occur, and quantifying their impact on profit.

- Increase overall profitability, not just revenue, by connecting order status, discounting behavior, and category mix to the bottom line.

4. Project Scope & Data Sources

The project draws on five related source datasets that together describe the full customer journey, from browsing to purchase to post-purchase review. These were combined into a single analytical table for exploration in Python and SQL, and a transaction-level extract was also modeled directly in Power BI for interactive reporting.

4.1 Source Tables

Dataset	Description	Key Fields
customers.csv	Customer master data used to profile who is buying.	customer_id, demographic attributes
products.csv	Product catalog with pricing, ratings, and stock information.	product_id, product_name, category, brand, price, avg_rating, num_ratings, stock_quantity, discount_pct, is_featured, weight_kg
transactions.csv	Order-level transactional records — the core fact table.	transaction_id, customer_id, product_id, transaction_date, quantity, unit_price, total_amount, discount_applied, status, payment_method, shipping_cost
sessions.csv	Website/app browsing session data supporting behavioral analysis.	session-level browsing attributes
reviews.csv	Customer product reviews supporting rating-based analysis.	review-level rating and text data

Table 1: Source datasets used in the analysis.

The transactions table is the central fact table: every row represents a single purchased line item, linked to a customer (customer_id) and a product (product_id). Products and transactions were merged on product_id to bring category, brand, price, and rating information onto every transaction, and customers were subsequently joined on customer_id to add demographic context. After merging, engineered fields such as revenue (quantity × price) were added to support category- and product-level financial analysis.

4.2 Tools and Technologies

- Python (pandas, NumPy) — data loading, cleaning, merging, and feature engineering.
- Matplotlib and Seaborn — exploratory visualizations (bar charts, pie charts, line plots).
- MySQL (via SQLAlchemy) — relational storage of the cleaned dataset and SQL-based business querying.
- Power BI — interactive dashboard for revenue, profit, order, and growth-trend reporting.
- Jupyter Notebook — primary analysis environment tying the Python workflow together.

5. Data Preparation & Methodology

Data preparation followed a standard, reproducible pipeline so that the same cleaned dataset could support both the Python/SQL analysis and the Power BI dashboard.

5.1 Loading and Merging

The five raw CSV files were loaded into pandas DataFrames. Because products and transactions initially shared only the `product_id` key, the transactions table was merged with the products table on `product_id` using a left join, preserving every transaction even where product metadata was incomplete. The result was then merged with the customers table on `customer_id` to bring in customer-level context. This produced a single working DataFrame containing transaction, product, and customer attributes in one place.

```
df = transactions.merge(products, on="product_id", how="left") df = df.merge(customers,
on="customer_id") df["revenue"] = df["quantity"] * df["price"]
```

5.2 Feature Engineering

- `revenue` — calculated as quantity multiplied by unit price, used as the primary financial metric throughout the Python and SQL analysis.
- `age_group` — customer ages were binned into descriptive groups (e.g., Young Adult, Adult, Middle-aged, Senior) to support demographic revenue analysis.
- `purchase_frequency_days` — derived from purchase timing to help distinguish one-time buyers from repeat customers.

5.3 Data Quality Checks

Null-value checks were run across every column of the merged dataset (`df.isnull().sum()`) and confirmed that, after the merge, there were no missing values in the fields used for analysis. Data types were also validated — for example, confirming that `product_id` was consistently typed as a string in both the products and transactions tables before merging, which is a common source of failed joins. A redundancy check was performed between the `discount_applied` and `promo_code_used` fields; since they carried overlapping information, `promo_code_used` was dropped in favor of the more directly useful `discount_applied` flag.

5.4 Loading into MySQL

Once cleaned, the merged DataFrame was written out and loaded into a MySQL database named `E_COMMERCE_BEHAVIOR`, into a table called `shopping_data`, using SQLAlchemy's `to_sql()` function. This step made the dataset queryable with standard SQL, which was used for the structured business-question analysis described in Section 7. The database connection credentials used during this step are excluded from this report for security reasons.

6. Exploratory Data Analysis (Python)

With the merged dataset in hand, a series of visual explorations were carried out to understand demand, revenue concentration, pricing behavior, and the relationship between product quality (ratings) and sales. Each chart below was generated directly from the cleaned transaction-level data.

6.1 Top 10 Products by Demand

The best-selling products by total units sold are led by “PureBrand Sports #70” and “FitGear Home & Garden #702”, both moving over 230 units. The top ten products span a wide mix of categories — Sports, Home & Garden, Electronics, Clothing, Software, Office Supplies, Automotive, and Food & Grocery — indicating that demand is not concentrated in a single category, even though revenue is (see Section 6.2). This distinction between unit demand and revenue concentration is an important input into merchandising and inventory decisions.

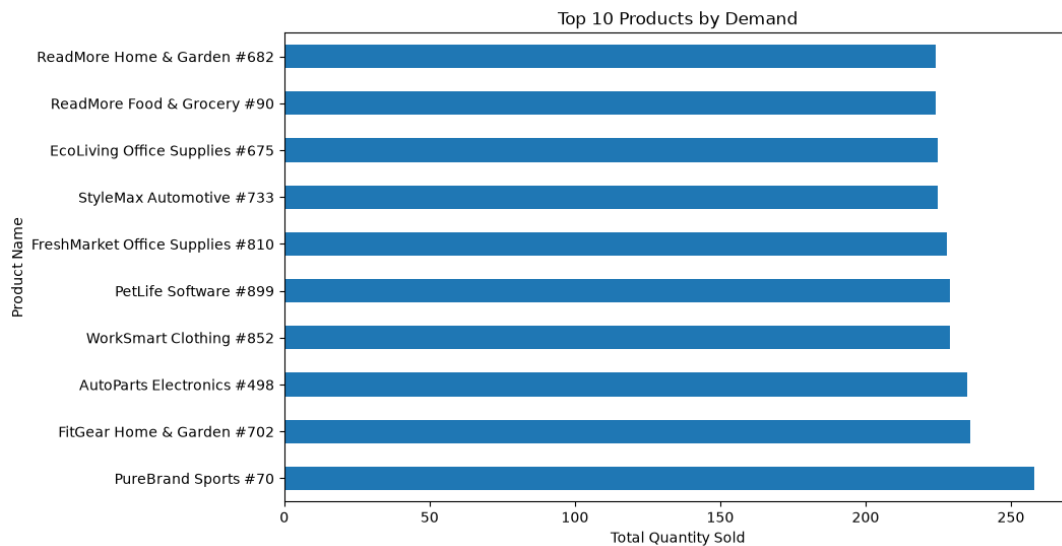


Figure 1: Top 10 products ranked by total quantity sold.

6.2 Revenue by Category

When products are aggregated by category, Electronics is decisively the top revenue generator, bringing in roughly \$2.7M — more than double the next closest category, Jewelry (around \$1.3M), followed by Automotive (around \$1.1M). Home & Garden, Sports, Health, and Clothing form a mid-tier cluster each contributing roughly \$0.6M–\$0.8M, while Books, Food & Grocery, and Music are the smallest contributors at under \$0.2M each. This confirms, at the SKU level, the same category concentration that the Power BI dashboard later reports at the order level.

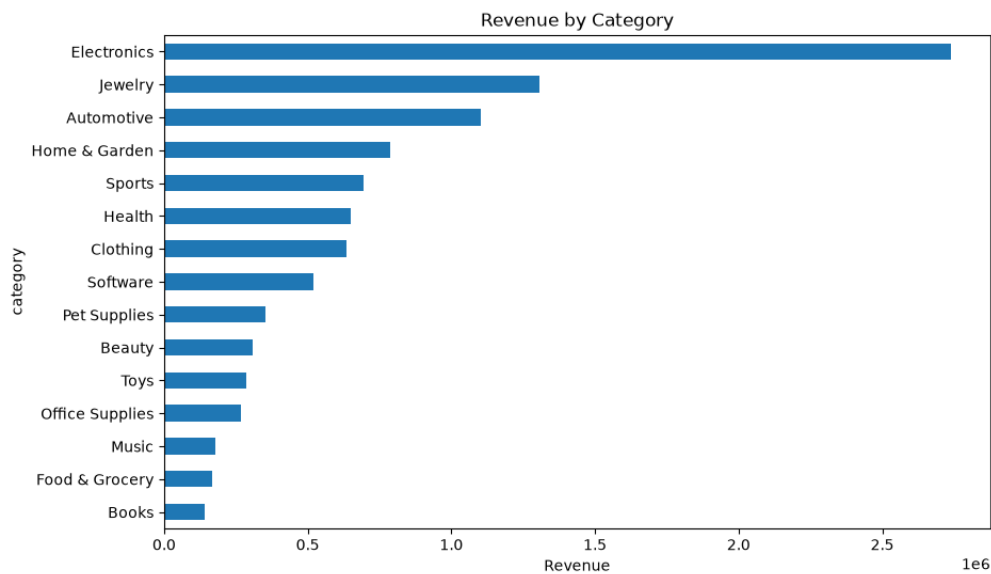


Figure 2: Total revenue by product category.

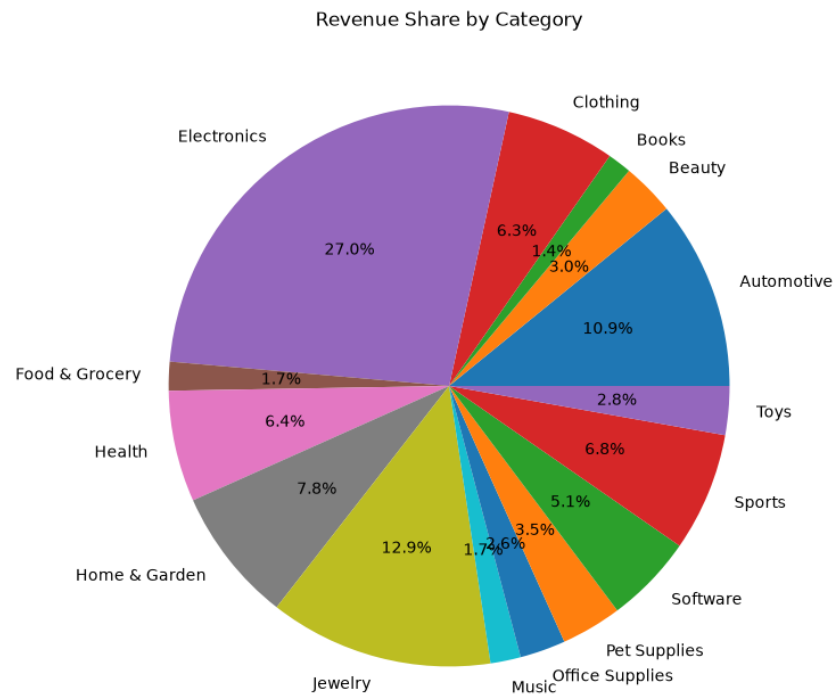


Figure 3: Revenue share by category — Electronics alone accounts for approximately 27% of total revenue, followed by Jewelry (12.9%) and Automotive (10.9%).

6.3 Revenue by Payment Method

Credit card is the dominant payment method by revenue, generating roughly \$3.5M, well ahead of debit card (~\$2.0M) and PayPal (~\$1.8M). Digital wallets — Apple Pay and Google Pay — each contribute just over \$1M, while bank transfer is the smallest channel at roughly \$0.5M. This distribution is useful for prioritizing which payment integrations and checkout experiences deserve the most investment, and for understanding where payment-related friction (e.g., failed transactions or cancellations) would have the largest revenue impact.

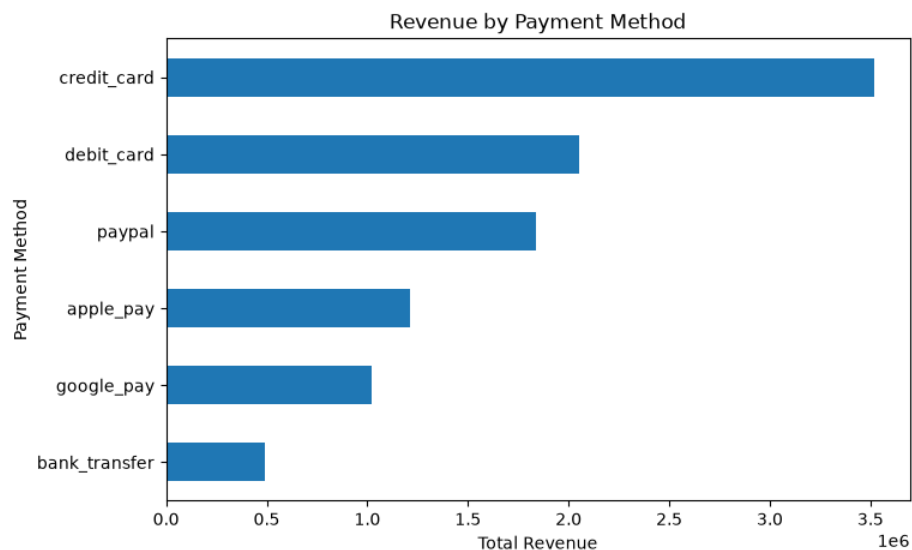


Figure 4: Total revenue by payment method.

6.4 Brand Performance

At the brand level, revenue is led by EcoLiving and NovaTech, each generating roughly \$850K, noticeably ahead of GlowUp, StyleMax, ClassicCo, PureBrand, ShineOn, WorkSmart, HomeEssentials, and AutoParts, which cluster more tightly in the \$480K–\$570K range. Among just the top five brands, EcoLiving (26.0%) and NovaTech (25.2%) together account for roughly half of that group's combined revenue, with GlowUp, StyleMax, and ClassicCo splitting the remainder relatively evenly.

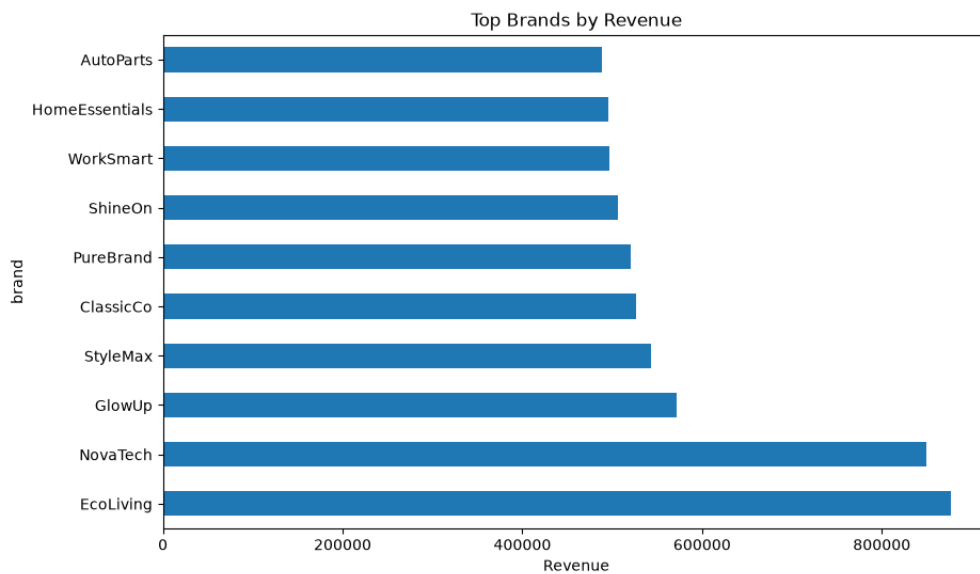


Figure 5: Top 10 brands by total revenue.

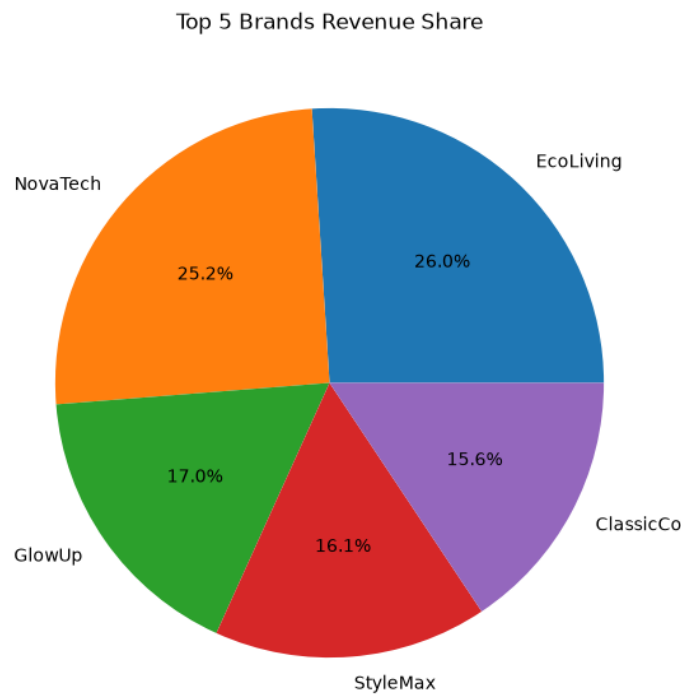


Figure 6: Revenue share among the top 5 brands.

6.5 Discount vs. Average Revenue

Plotting average revenue against discount percentage reveals a non-linear, somewhat volatile relationship rather than a simple “bigger discount, bigger revenue” pattern. Average revenue actually peaks around a 10% discount level (approximately \$99) and again around 40% (approximately \$108), but drops sharply at 25% (down to roughly \$67) — its lowest point on the chart.

This suggests that discount depth alone does not reliably predict revenue outcomes, and that specific discount tiers may be interacting with product mix or promotional timing in ways that merit closer investigation before broad discount policy changes are made.

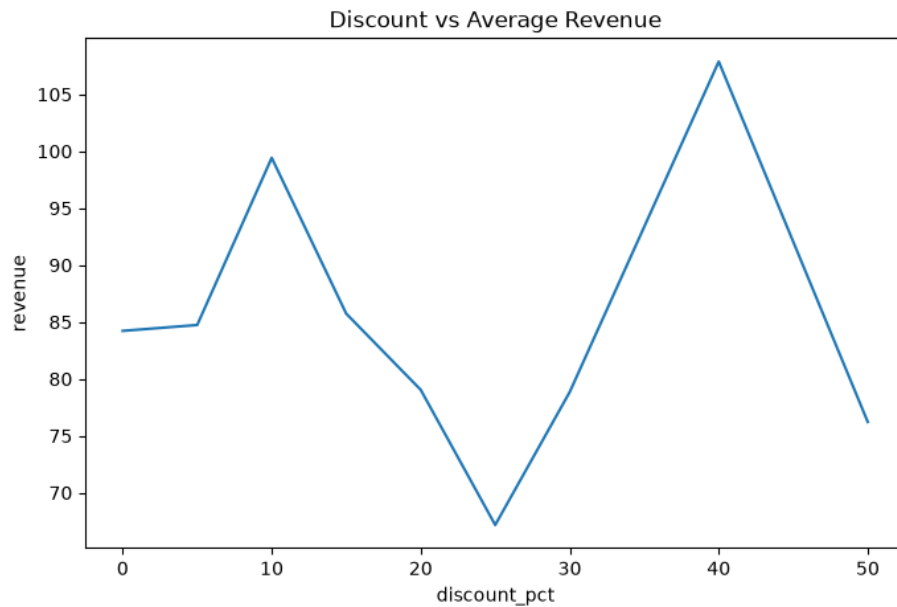


Figure 7: Relationship between discount percentage and average revenue.

6.6 Product Rating vs. Demand

There is a broadly positive relationship between average product rating and the quantity sold: products rated below roughly 3.0 sell in very small volumes (under ~2,000 units in aggregate), while quantity climbs substantially for products rated between 3.0 and 4.5, and spikes sharply for the small number of products rated at or near 5.0 (approaching 16,500 units). This supports the intuitive but important conclusion that product quality, or at least perceived quality via ratings, is one of the strongest visible levers on demand, and reinforces the value of actively managing and improving low-rated listings.

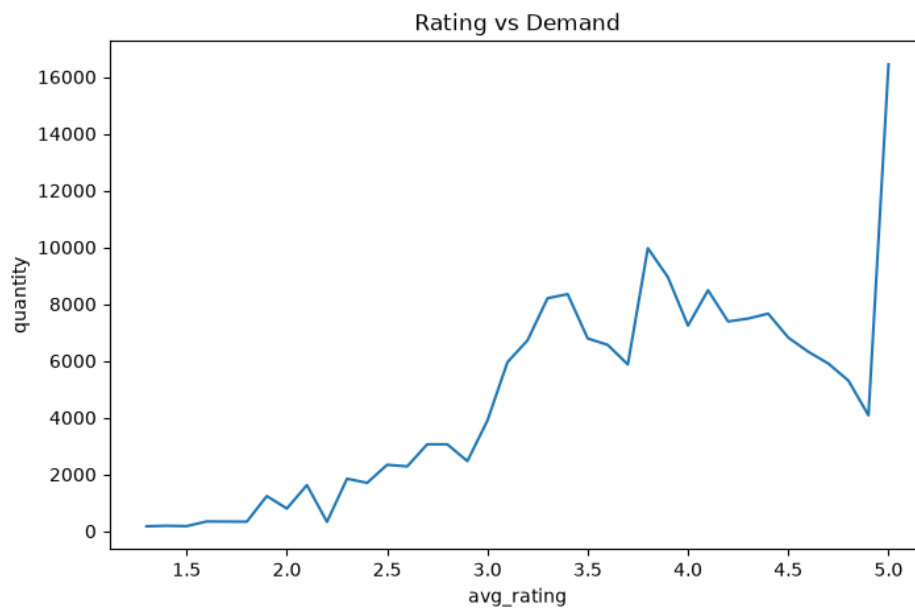


Figure 8: Relationship between average product rating and total quantity sold.

7. SQL-Based Business Analysis

After the cleaned dataset was loaded into the shopping_data table in the E_COMMERCE_BEHAVIOR MySQL database, fourteen structured SQL queries were written and executed to answer specific, recurring business questions. Each query below is presented with its purpose and the underlying SQL logic; together they form a repeatable analytical toolkit that the business can re-run as new data arrives.

Q1 — Total Revenue Generated

Sums the revenue field across every transaction to give a single top-line revenue figure for the business.

```
SELECT CONCAT('$', ROUND(SUM(revenue))) AS Total_Revenue FROM shopping_data;
```

Q2 — Total Orders Placed

Counts every row in the transaction table to quantify total order volume.

```
SELECT COUNT(*) AS Total_Orders FROM shopping_data;
```

Q3 — Unique Customers

Counts distinct customer IDs to measure the size of the active customer base.

```
SELECT COUNT(DISTINCT customer_id) AS Unique_Customers FROM shopping_data;
```

Q4 — Most Demanded Products

Ranks products by total quantity sold to identify the top 15 products by unit demand.

```
SELECT product_name, SUM(quantity) AS Total_quantity FROM shopping_data GROUP BY product_name ORDER BY total_quantity DESC LIMIT 15;
```

Q5 — Revenue by Category

Aggregates revenue by product category to see which categories generate the most money, not just the most orders.

```
SELECT category, CONCAT('$', ROUND(SUM(revenue),0)) AS Total_Most_Revenue FROM shopping_data GROUP BY category ORDER BY total_most_revenue DESC;
```

Q6 — Category Ratings

Computes the average product rating per category to flag categories with quality or satisfaction issues.

```
SELECT category, ROUND(AVG(avg_rating),1) AS avg_rating FROM shopping_data GROUP BY category ORDER BY avg_rating;
```

Q7 — Highest-Revenue Products

Identifies the individual products (rather than categories) generating the most revenue.

```
SELECT product_name, ROUND(SUM(revenue),2) AS total_revenue FROM shopping_data GROUP BY product_name ORDER BY total_revenue DESC LIMIT 10;
```

Q8 — Monthly Sales Trend

Groups revenue by calendar month to reveal the seasonal pattern referenced throughout this report.

```
SELECT MONTH(transaction_date) AS month, CONCAT('$', ROUND(SUM(revenue),2)) AS month_revenue FROM
shopping_data GROUP BY month ORDER BY month;
```

Q9 — Above-Average Revenue Products

Uses a HAVING clause against the overall average revenue to surface products that outperform the typical product.

```
SELECT product_name, ROUND(SUM(revenue),2) AS total_revenue FROM shopping_data GROUP BY product_name
HAVING total_revenue > (SELECT AVG(revenue) FROM shopping_data);
```

Q10 — Top Revenue Category Drill-Down

Uses a subquery to find the single highest-revenue category, then lists every product within it.

```
SELECT product_name, category FROM shopping_data WHERE category = ( SELECT category FROM
shopping_data GROUP BY category ORDER BY SUM(revenue) DESC LIMIT 1 );
```

Q11 — Buyers of the Most Expensive Product

Finds which customers purchased the single highest-priced product in the catalog, useful for premium-segment targeting.

```
SELECT DISTINCT customer_id, product_name FROM shopping_data WHERE price = (SELECT MAX(price) FROM
shopping_data);
```

Q12 — Top 3 Products per Category

Uses a window function (RANK) partitioned by category to surface each category's top three products by revenue — the same logic behind the sample report's category leaderboard.

```
SELECT * FROM ( SELECT product_name, category, ROUND(SUM(revenue),2) AS total_revenue, RANK()
OVER (PARTITION BY category ORDER BY SUM(revenue) DESC) AS rank_position FROM shopping_data GROUP
BY product_name, category ) ranked WHERE rank_position <= 3;
```

Q13 — Repeat Customers

Counts transactions per customer and filters to those with more than one purchase, forming the basis of repeat-customer and loyalty analysis.

```
SELECT customer_id, COUNT(transaction_id) AS purchase_count FROM shopping_data GROUP BY customer_id
HAVING purchase_count > 1;
```

Q14 — Highest-Revenue Day

Uses a window function to rank every calendar date by total revenue and returns the single best-performing day.

```
SELECT * FROM ( SELECT transaction_date, SUM(revenue) AS daily_revenue, RANK() OVER (ORDER BY
SUM(revenue) DESC) AS rank_position FROM shopping_data GROUP BY transaction_date ) ranked WHERE
rank_position = 1;
```

8. Power BI Dashboard

To make the findings accessible to non-technical stakeholders and support ongoing monitoring, an interactive Power BI dashboard was built on top of the transaction-level data. The dashboard lets users filter by product category, year, and order status, and surfaces the metrics that matter most for tracking the business problem described in Section 2: revenue, profit, order volume, and growth.

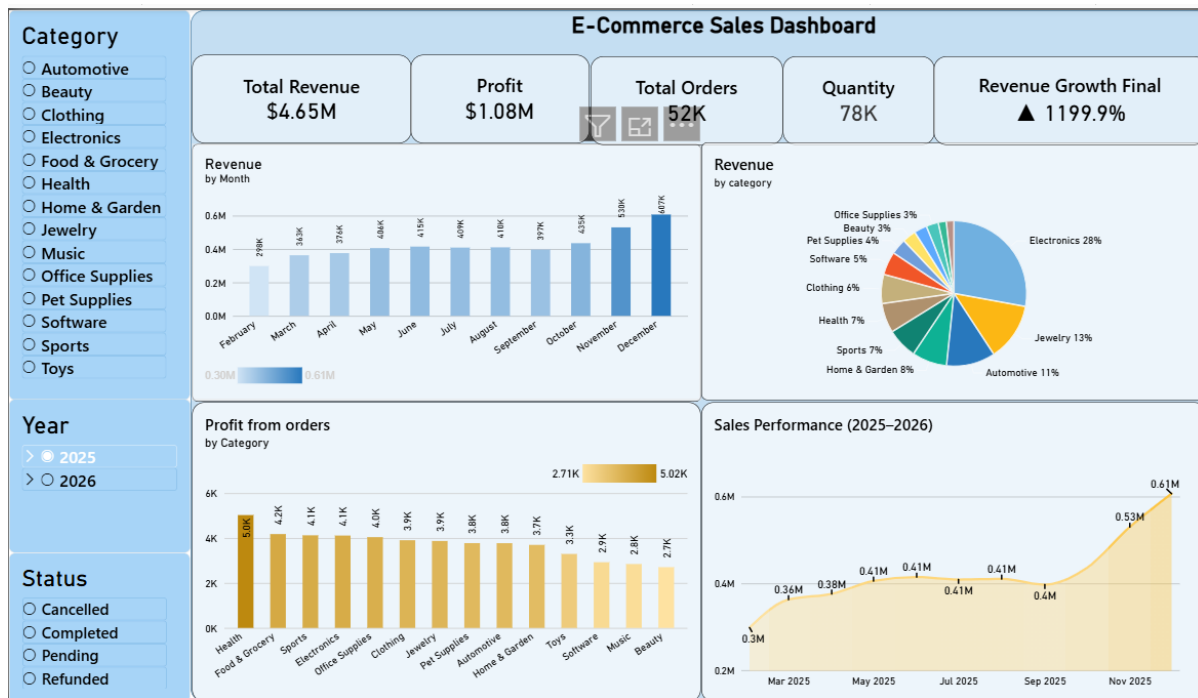


Figure 9: E-Commerce Sales Dashboard (Power BI) — headline KPIs, monthly revenue, category mix, profit by category, and 2025–2026 sales performance.

8.1 Headline KPIs



The dashboard is organized around five visual panels. The Revenue by Month bar chart shows a clear upward seasonal trend, rising from roughly \$298K in February to a peak of \$607K in December, with a dip visible around September–October before the year-end recovery. The Revenue by Category donut chart mirrors the Python analysis, showing Electronics as the leading category (28%), followed by Jewelry (13%) and Automotive (11%), with the remaining categories — Home & Garden, Sports, Health, Clothing, Software, Pet Supplies, Beauty, and Office Supplies — each contributing single-digit percentages.

The Profit from Orders by Category chart tells a materially different story from the revenue chart: Health leads on profit (5.0K), followed by Food & Grocery, Sports, Electronics, and Office Supplies, while Beauty, Music, and Toys sit at the bottom. This is the clearest evidence in the entire project that high revenue and high profit are not the same thing — Electronics generates the most revenue but is not the top profit contributor, which directly supports the business objective of optimizing category performance rather than simply chasing top-line sales. Finally, the Sales Performance (2025–2026) area chart shows a relatively flat trend through most of 2025 (roughly \$0.3M–\$0.4M) before accelerating sharply from around October 2025 onward, reaching \$0.61M by the most recent period shown, consistent with the 1199.9% year-over-year growth figure in the headline KPIs.

8.2 Filters and Interactivity

The left-hand navigation panel allows users to slice every visual by Category (15 categories from Automotive through Toys), by Year (2025 and 2026), and by order Status (Cancelled, Completed, Pending, Refunded). This last filter is particularly

important given the business problem: it allows stakeholders to isolate the revenue and profit impact of Cancelled and Refunded orders specifically, rather than only seeing blended totals that can mask how much value is being lost to order failures.

9. Key Insights & Findings

Bringing together the Python/SQL exploratory analysis and the Power BI dashboard, the following insights stand out as the most decision-relevant findings from this project:

- Revenue is strongly seasonal, with a clear trough in February and a strong build into Q4, peaking in December. Any forecasting or staffing model should treat this pattern as a baseline rather than an anomaly.
- Electronics is the dominant revenue category (roughly 27–28% of total revenue across both the SKU-level and order-level views), but it is not the leading profit category — Health, Food & Grocery, and Sports convert more efficiently into profit.
- Revenue and profit must be tracked separately. Orders and quantity sold do not move in lockstep with profit, so category and product decisions based on revenue alone risk under-valuing genuinely profitable segments.
- Cancelled and Refunded orders are a direct and quantifiable drag on profitability. Because the dashboard's Status filter isolates these order types, the business can now measure their financial impact on demand rather than estimating it.
- Demand is broad-based across categories, but revenue and brand value are concentrated. The top 10 products by unit volume span eight different categories, while revenue is led overwhelmingly by a small number of categories and brands (EcoLiving and NovaTech alone account for roughly half of the top-five brands' combined revenue).
- Discount depth has a volatile, non-linear relationship with average revenue, with a notable dip at the 25% discount level. Blanket discounting strategies should be re-evaluated in favor of tier-specific testing.
- Product rating is one of the strongest visible drivers of demand, with near-5-star products selling in dramatically higher volumes than products rated below 3.0. Review and rating management is therefore a demand lever, not just a customer-satisfaction metric.
- Credit card is the leading payment method by a wide margin, followed by debit card and PayPal, information that is useful for prioritizing checkout and payment-reliability investments.

10. Business Recommendations

10.1 Reduce Order Cancellations

Use the Status filter in the dashboard to build a recurring (e.g., monthly) cancellation and refund report broken out by category and payment method, to identify whether cancellations cluster around specific products, price points, shipping options, or payment types. Root-causing even the top two or three cancellation drivers is likely to have an outsized effect on the \$1.08M profit base, since cancelled orders still consume fulfillment resources without generating retained revenue.

10.2 Rebalance Category Investment Toward Profit, Not Just Revenue

Because Electronics leads on revenue but not on profit, marketing and promotional spend should not be allocated purely in proportion to revenue share. Health, Food & Grocery, and Sports — the strongest profit contributors — deserve increased visibility (homepage placement, bundling, targeted promotions) even though they generate less absolute revenue than Electronics.

10.3 Re-Test Discount Tiers Rather Than Broad Discounting

Given the volatile relationship between discount percentage and average revenue — including a clear dip at 25% — the business should run controlled A/B tests at the 10%, 25%, and 40% discount levels on comparable products to understand causally why the 25% tier underperforms, rather than assuming deeper discounts are always better or worse.

10.4 Invest in Ratings and Review Quality

Because demand rises sharply with rating, and especially for products approaching 5.0, the business should prioritize operational levers that protect and improve ratings: proactive quality control for lower-rated SKUs, incentivized post-purchase reviews for high-potential new products, and de-prioritizing or improving products that persistently sit below a 3.0 average rating.

10.5 Plan Inventory and Staffing Around the Q4 Seasonal Peak

With revenue climbing from roughly \$298K in February to \$607K in December, and the 2025–2026 trend line showing a sharp acceleration from October onward, inventory buffers, warehouse staffing, and customer support capacity should be scaled up ahead of Q4 rather than reactively, and the February trough should be used as a window for clearance promotions, maintenance, and process improvement work.

10.6 Concentrate Brand and Vendor Negotiations on Top Performers

Since EcoLiving and NovaTech together represent roughly half of the combined revenue of the top five brands, the business is well positioned to negotiate improved vendor terms, exclusive product lines, or co-marketing arrangements with these partners, while using the broader brand data to identify which mid-tier brands (e.g., GlowUp, StyleMax, ClassicCo) have room to grow with additional support.

10.7 Continue Operationalizing the Power BI Dashboard

The dashboard should be adopted as a living operational tool rather than a one-time deliverable: refreshed on a regular cadence, extended with a cancellation-reason breakdown if that data becomes available, and reviewed in a recurring business meeting so that the category, status, and year filters are actively used to catch emerging revenue or profit issues early.

11. Conclusion

This project combined Python-based exploratory analysis, SQL-based business querying, and an interactive Power BI dashboard to address a clearly defined business problem: inconsistent revenue growth, category imbalance, and profit loss from cancelled orders. The analysis confirms that Electronics leads on revenue but not on profit, that revenue is strongly seasonal with a Q4 peak and a February trough, that discounting behaves non-linearly rather than predictably, and that product ratings are a meaningful demand driver. Cancelled and refunded orders are now directly measurable through the dashboard's status filter, closing the loop between the original business problem and an actionable, ongoing monitoring tool.

Taken together, the recommendations in Section 10 give the business a concrete path toward the five objectives set out at the start of the project: more consistent revenue growth, clear visibility into high- and low-performing periods, better-optimized category investment, fewer cancellations, and — ultimately — higher overall profitability. The underlying SQL toolkit and Power BI model are both designed to be re-run as new transactional data arrives, so this report should be treated as a baseline rather than a one-off snapshot.